THE CENTERED COLLISION SUM

ALEXANDER S. PETTY

Abstract. For a prime $p > b$ and lag $\ell \geq 1$, the collision deviation $S_\ell(p) = C(b^\ell \bmod p) - Q$ depends only on $p \bmod b^{\ell+1}$ (the finite determination theorem). This reduces the collision deviation to a function on the finite group $(\mathbb{Z}/b^{\ell+1}\mathbb{Z})^\times$.

The *collision transform* decomposes this function over Dirichlet characters modulo $b^{\ell+1}$. The trivial coefficient is exactly $-1/2$, forced by the reflection identity $S(a) + S(m-a) = -1$. The non-trivial characters carry the centered fluctuation.

Subtracting the per-class mean defines the centered fluctuation sum $F^\circ(1,\ell) = \sum_p S_p^\circ/p$. By Mertens' theorem for arithmetic progressions, F° converges at $s = 1$ for every base b and lag ℓ. This is the Centered Collision PNT: the centered collision fluctuations cancel at the edge of the critical strip.

For prime bases at lag 1, finite evaluation suggests the explicit class mean formula $\overline{S}_R = (R+1)/b - 1$, compatible with grand mean $-1/2$ and reflection symmetry $\overline{S}_R + \overline{S}_{b-2-R} = -1$.

1. Introduction

This paper proves the convergence of the centered collision fluctuation sum. The preceding paper [2] conjectured that the raw fluctuation sum $F(1,\ell)$ diverges at the Mertens rate $-\mu_\ell \cdot \log\log x$, driven by a persistent per-class bias. The present paper identifies the bias exactly, subtracts it, and proves that the remainder converges.

The proof has three steps. First, the finite determination theorem (Section 2) shows that $S_\ell(p)$ depends only on $p \bmod b^{\ell+1}$. Second, the collision transform (Section 3) decomposes the resulting finite function over Dirichlet characters. Third, the Centered Collision PNT (Section 6) assembles the convergence from Mertens' theorem for arithmetic progressions.

2. The Finite Determination Theorem

Theorem 1 (Finite determination)**.** *For any base $b \geq 2$, lag $\ell \geq 1$, and integer $p > b^{\ell+1}$ with $\gcd(p,b) = 1$, the collision deviation $S_\ell(p) =$*

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$C(b^\ell \bmod p) - \lfloor (p-1)/b \rfloor$ *depends only on* $p \bmod b^{\ell+1}$. *Primality of* p *is not required.*

Proof. Set $m = b^{\ell+1}$. For each $r \in \{1, \ldots, p-1\}$, define the slice index $n(r) = \lfloor mr/p \rfloor \in \{0, \ldots, m-1\}$.

Since the slice width $p/m > 1$, each slice contains integers. The digit function and the collision multiplier are constant on each slice:

$$\delta(r) = \left\lfloor \frac{br}{p} \right\rfloor = \left\lfloor \frac{n}{b^\ell} \right\rfloor, \qquad \delta(b^\ell r \bmod p) = n \bmod b.$$

A slice is *diagonal* if $\lfloor n/b^\ell \rfloor = n \bmod b$. Let $G_{\ell,b} = \{n \in \{0, \ldots, m-1\} : \lfloor n/b^\ell \rfloor = n \bmod b\}$. Then $|G_{\ell,b}| = b^\ell$ and

$$C(b^\ell \bmod p) = -1 + \sum_{n \in G_{\ell,b}} \left(\left\lfloor \frac{(n+1)p}{m} \right\rfloor - \left\lfloor \frac{np}{m} \right\rfloor \right).$$

Write $p = mt + a$ with $1 \le a < m$ and $\gcd(a, b) = 1$. Then

$$\left\lfloor \frac{(n+1)p}{m} \right\rfloor - \left\lfloor \frac{np}{m} \right\rfloor = t + \left\lfloor \frac{(n+1)a}{m} \right\rfloor - \left\lfloor \frac{na}{m} \right\rfloor.$$

Since $Q = b^\ell t + \lfloor a/b \rfloor$:

$$S_\ell(p) = -1 - \left\lfloor \frac{a}{b} \right\rfloor + \sum_{n \in G_{\ell,b}} \left(\left\lfloor \frac{(n+1)a}{m} \right\rfloor - \left\lfloor \frac{na}{m} \right\rfloor \right),$$

which depends only on $a = p \bmod m$. □

Remark 2. The modulus $b^{\ell+1}$ is sharp. In base 10 at lag 2: $S_2(1009) = 8$ and $S_2(1109) = -10$, though $1009 \equiv 1109 \pmod{100}$. Reduction modulo b^2 does not determine S_2; modulo $b^3 = 1000$ does.

3. The Collision Transform

Since S_ℓ is a function on $(\mathbb{Z}/b^{\ell+1}\mathbb{Z})^\times$, it admits a Fourier expansion over Dirichlet characters.

Definition 3. The *collision transform* is

$$\hat{S}_\ell(\chi) = \frac{1}{\phi(b^{\ell+1})} \sum_{a \in (\mathbb{Z}/b^{\ell+1}\mathbb{Z})^\times} S_\ell(a)\, \overline{\chi}(a)$$

for each Dirichlet character χ modulo $b^{\ell+1}$.

Theorem 4 (Reflection identity)**.** *For* $m = b^{\ell+1}$ *and any unit* a*:*

$$S_\ell(a) + S_\ell(m - a) = -1.$$

Proof. Replacing a by $m-a$ in the formula of Theorem 1 sends $\lfloor na/m \rfloor$ to $n-1-\lfloor na/m \rfloor$ for $0 < na < m$ (the complementary floor identity). The sum over diagonal slices transforms accordingly, giving $S_\ell(m-a) = -1 - S_\ell(a)$. □

Corollary 5 (Trivial coefficient)**.** *$\hat{S}_\ell(\chi_0) = -1/2$.*

Proof. The involution $a \mapsto m-a$ pairs the units. By Theorem 4, each pair averages to $-1/2$. □

Corollary 6 (Nontrivial even characters vanish)**.** *For every nontrivial even character χ (i.e., $\chi(-1) = +1$): $\hat{S}_\ell(\chi) = 0$.*

Proof. Since χ is even, $\chi(-a) = \chi(a)$. Pairing a with $-a$ and using Theorem 4,

$$\hat{S}_\ell(\chi) = \frac{1}{2\phi(b^{\ell+1})} \sum_a \big(S_\ell(a) + S_\ell(-a)\big)\overline{\chi}(a) = -\frac{1}{2\phi(b^{\ell+1})} \sum_a \overline{\chi}(a).$$

For a nontrivial character, the last sum is zero. □

4. The Half-Group Structure

Theorem 7 (Half-group)**.** *For every diagonal slice $n \in G_{\ell,b}$ with $n \neq m-1$, the wrapping set $W_n = \{a : (n+1)a \bmod m < a\}$ satisfies $|W_n| = \phi(m)/2$.*

Proof. Let $c = n+1$. If $a \in W_n$, then $c(m-a) \bmod m = m - (ca \bmod m) > m-a$, so $m-a \notin W_n$. The involution $a \mapsto m-a$ maps W_n bijectively onto its complement. □

5. The Class Mean Formula

Definition 8. For spectral class $R = (p-1) \bmod b$, the class mean is

$$\overline{S}_R = \frac{1}{|\mathcal{A}_R|} \sum_{a \in \mathcal{A}_R} S_\ell(a),$$

where $\mathcal{A}_R = \{a \in (\mathbb{Z}/b^{\ell+1}\mathbb{Z})^\times : a \equiv R+1 \pmod{b}\}$.

Observation 9 (Prime-base class mean at lag 1)**.** *For prime base b at lag $\ell = 1$: $\overline{S}_R = (R+1)/b - 1$.*

This is obtained by evaluating $S_1(a)$ at all $a \in (\mathbb{Z}/b^2\mathbb{Z})^\times$ using Theorem 1 and averaging over each class.

For composite bases, the class means are rational and satisfy the following universal properties.

Corollary 10. *For all bases $b \geq 2$ at any lag $\ell \geq 1$, let $\mathcal{R} = \{R \in \{0, \ldots, b-2\} : \gcd(R+1, b) = 1\}$ be the set of admissible spectral classes. Then:*

(i) *Grand mean:* $\dfrac{1}{|\mathcal{R}|} \displaystyle\sum_{R \in \mathcal{R}} \overline{S}_R = -\frac{1}{2}$.

(ii) *Reflection symmetry:* $\overline{S}_R + \overline{S}_{b-2-R} = -1$ *for every* $R \in \mathcal{R}$.

Proof. Property (i) follows from Corollary 5. Property (ii) follows from the reflection identity: the map $a \mapsto m - a$ sends class R to class $b-2-R$. □

R	0	2	6	8
$\overline{S}_R$ (exact)	$-17/10$	$-9/10$	$-1/10$	$7/10$
computed	-1.709	-0.897	-0.097	$+0.699$

TABLE 1. Class means in base 10 at lag 1, computed over 166,000 primes per class (664,574 total, $p \leq 10^7$).

6. The Centered Collision PNT

Definition 11. The centered collision deviation is $S_p^\circ = S_\ell(p) - \overline{S}_{R(p)}$. The centered fluctuation sum is

$$F^\circ(s, \ell) = \sum_{\substack{p > b^{\ell+1} \\ \gcd(p,b)=1}} \frac{S_p^\circ}{p^s}.$$

Theorem 12 (Centered Collision PNT)**.** *For every base $b \geq 2$ and lag $\ell \geq 1$, $F^\circ(1, \ell)$ converges.*

Proof. Set $q = b^{\ell+1}$. By Theorem 1, $S_\ell(p)$ depends only on the residue class of p modulo q, so the same is true for the centered quantity S_p°. Define the finite function

$$f^\circ(a) = S_\ell(a) - \overline{S}_{R(a)} \qquad \big(a \in (\mathbb{Z}/q\mathbb{Z})^\times\big),$$

where $R(a)$ is determined by $a \equiv R(a) + 1 \pmod{b}$. By definition of the class means,

$$\sum_{a \in \mathcal{A}_R} f^\circ(a) = 0$$

for every spectral class $\mathcal{A}_R$, hence in particular

$$\sum_{a \in (\mathbb{Z}/q\mathbb{Z})^\times} f^\circ(a) = 0.$$

Therefore the trivial Fourier coefficient vanishes, and the collision transform expands as

$$f^\circ(a) = \sum_{\chi \neq \chi_0} \hat{f}^\circ(\chi)\, \chi(a),$$

with a finite sum over nontrivial Dirichlet characters modulo q. Substituting $a = p \bmod q$ gives

$$F^\circ(1, \ell) = \sum_{\chi \neq \chi_0} \hat{f}^\circ(\chi) \sum_{\substack{p > b^{\ell+1} \\ p \nmid b}} \frac{\chi(p)}{p}.$$

For each nontrivial χ modulo q, Mertens' theorem for arithmetic progressions (see, for example, the discussion in [5, Ch. 22]) gives

$$\sum_{\substack{p \leq x \\ p \equiv a \pmod q}} \frac{1}{p} = \frac{\log\log x}{\phi(q)} + M_a + o(1).$$

Summing this identity against $\chi(a)$ over the reduced residue classes modulo q yields

$$\sum_{p \leq x} \frac{\chi(p)}{p} \;=\; \frac{\log\log x}{\phi(q)} \sum_a \chi(a) + \sum_a \chi(a) M_a + o(1).$$

Because χ is nontrivial, $\sum_a \chi(a) = 0$, so the $\log\log x$ term cancels. Hence each prime sum

$$\sum_p \frac{\chi(p)}{p}$$

converges. Since only finitely many characters occur, the outer sum is finite, and therefore $F^\circ(1, \ell)$ converges. □

primes	$p_{\max}$	$F(1,1)$	$F(1,1)/\log\log p$	F°	$F^\circ/\log\log p$
1,000	7,949	−1.46	−0.664	0.020	0.009
10,000	104,773	−1.69	−0.690	0.023	0.009
100,000	1,299,791	−1.86	−0.705	0.023	0.009
664,574	9,999,991	−1.99	−0.714	0.023	0.008

TABLE 2. Raw vs. centered fluctuation sums in base 10 at lag 1. The raw ratio drifts toward a Mertens constant. The centered ratio decreases toward zero, consistent with convergence.

base	lag	primes	$F(1,\ell)$	F°	behavior
3	1	18,000	-1.59	0.163	small
7	1	42,000	-1.66	0.031	small
10	1	78,000	-1.76	0.043	small
10	2	78,000	-1.83	0.072	small
10	3	78,000	-0.56	-0.227	small
12	1	42,000	-1.67	0.129	small

TABLE 3. Centered fluctuation sums across bases and lags. All centered sums remain comparatively small over the computed range.

7. Computational Universality Checks

8. Computational Resonance Data

Define $\psi_k(p) = e(kc_p/p)$, where $c_p = b(1-b^\ell)^{-1} \bmod p$ is the collision parameter. The partial sums $\Sigma_k(x) = \sum_{p\le x} \psi_k(p)$ exhibit a dichotomy controlled by $\gcd(k, b-1)$.

k	$\gcd(k,b-1)$	$\lvert\Sigma_k\rvert/\sqrt{\pi(x)}$	behavior
1	1	0.16	cancels
2	1	0.21	cancels
3	3	140	resonates
5	1	0.06	cancels
7	1	0.22	cancels
9	9	280	resonates
10	1	0.15	cancels

TABLE 4. Fourier modes in base 10 at 78,493 primes. Modes with $\gcd(k, b-1) = 1$ are numerically small at this scale, while modes with $\gcd(k, b-1) > 1$ are numerically large.

The data suggest that the resonant modes correspond to characters modulo $b^{\ell+1}$ that factor through characters modulo b, while the non-resonant modes distinguish sub-classes within each spectral class.

9. Remarks

Connection to the alignment limit. The class mean $\overline{S}_0 = -2/3$ in base 3 is the alignment limit of the prime 3 from [1]. The constant 2/3

appears both as the alignment limit and as the magnitude of the class bias in the simplest nontrivial case. More generally, for prime base b, $\overline{S}_0 = 1/b - 1 = -(b-1)/b$.

The Mertens constant. The grand mean $-1/2$ controls the leading-order Mertens constant. By Mertens' theorem for arithmetic progressions, the raw sum diverges at rate $(-1/2) \cdot \log\log x$ from the grand mean alone. The per-class deviations from $-1/2$ and the spectral class corrections determine the exact Mertens constant μ_ℓ.

Toward an explicit collision PNT. A genuine collision analogue of the prime number theorem would give an asymptotic formula for the unweighted sum $\sum_{p \le x} S^\circ(p)$ with a proved error bound. The collision transform expresses S° as a linear combination of Dirichlet characters, and the classical explicit formula for character sums provides the machinery.

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Email address: `alexander.petty@gmail.com`